Curs 4

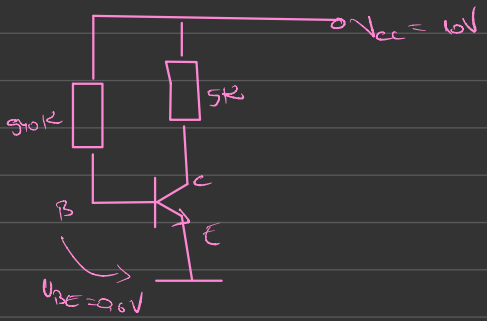
E.A. ⇒ M.F. ≈ M.E. ⇒ T. RAN

(Mărimi fizice) (Mărimi electronice)

Polarizare

$\begin{cases} I_C = \beta I_B \\ U_{BE} \simeq 0{,}6V \end{cases}$

I_C, U_{CE}

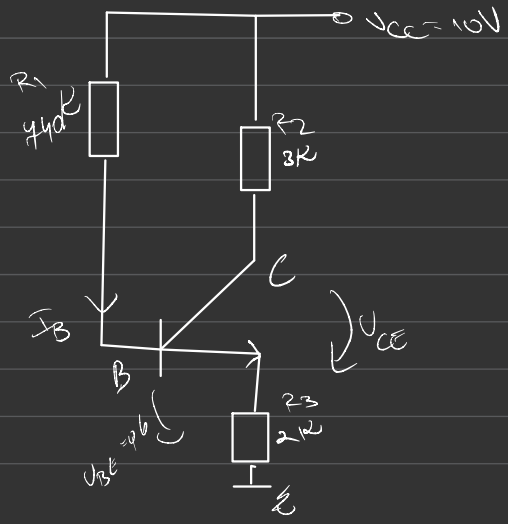


$I_B = 0{,}01\ mA$

$\beta = 100 \Rightarrow I_C = 1\ mA$, $U_{CE} = 5V$ — RAN

$\beta \geq 200 \Rightarrow I_C = 2\ mA$, $U_{CE} = 0V$ — ~~RAN~~

⇒ Mai adăugăm o rezistență:



$V_{CC} = R_1 I_B + U_{BE} + I_E R_3$

$I_B = \frac{I_C}{\beta}$

$I_E = I_B + I_C = \frac{I_C}{\beta} + I_C =$

$= I_C\left(\frac{1}{\beta} + 1\right) \simeq I_C$

REACȚIE NEGATIVĂ

$$I_C = \frac{V_{CC} - U_{BE}}{R_3 + \frac{R_1}{\beta}}$$

Für $\beta = 100 \Rightarrow I_C = \frac{9{,}4\,V}{7{,}4 + 2{,}2} = 1\,mA$

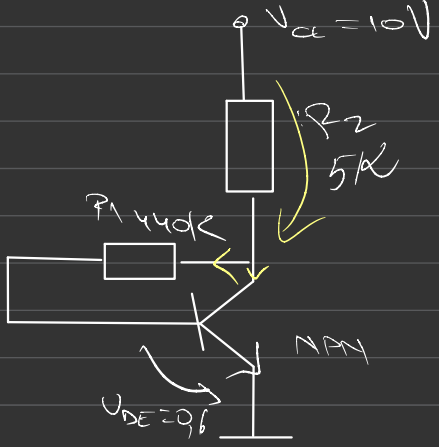
$$U_{CE} = V_{CC} - I_C(R_2 + R_3) = 10 - 5 = 5\,V$$

$$V_{CC} = I_C R_2 + U_{CE} + I_B R_3$$

$$\approx U_{CE} + I_C(R_2 + R_3)$$

Für $\beta = 200 \Rightarrow I_C = \frac{9{,}4\,V}{2 + 3{,}7} = 1{,}64\,mA$

$$U_{CE} = 10 - 5 - 1{,}64 = 1{,}8\,V$$



$$V_{cc} = R_2 (I_C + I_B) + R_1 I_B + V_{BE}$$

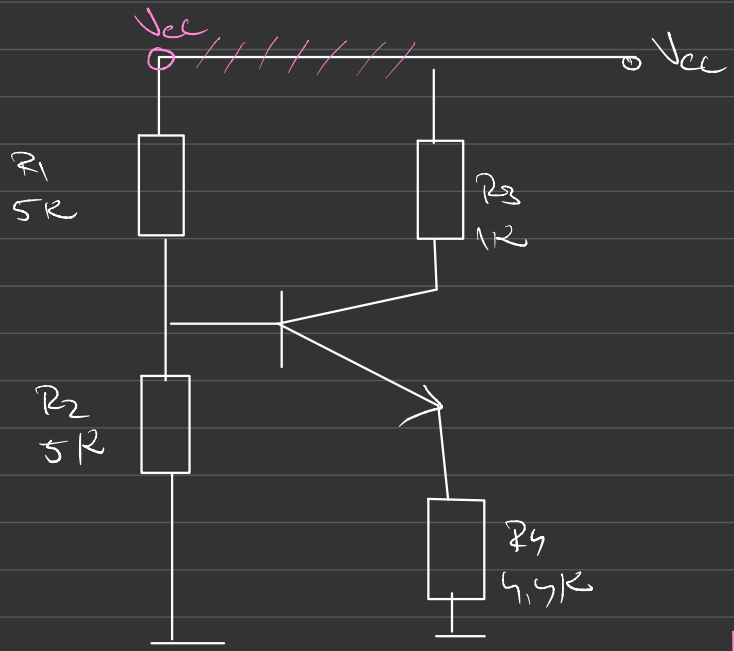
$$I_C = \frac{V_{cc} - V_{BE}}{R_2 + \frac{R_1}{\beta} + \frac{R_2}{\beta}} =$$

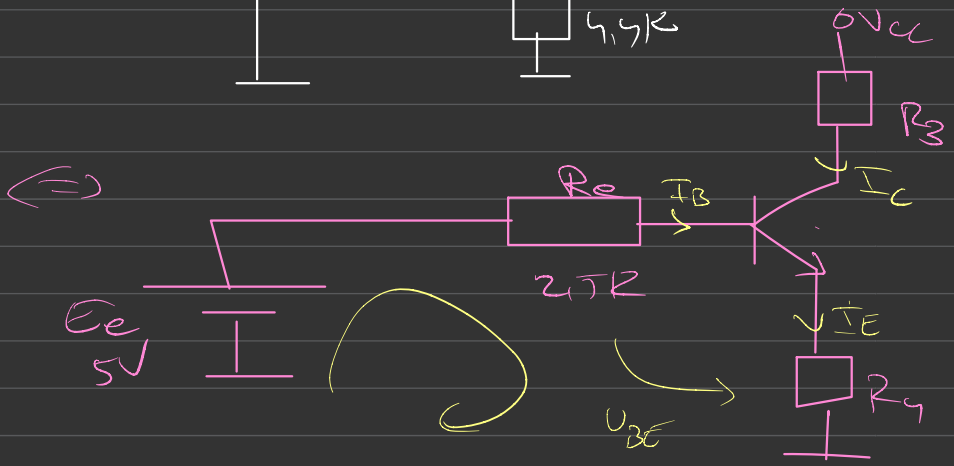
$$= \frac{9,4V}{5K + \frac{5K}{100} + \frac{440K}{100}} = \frac{9,4}{9,55} \simeq 1mA$$

$\beta = 100 \Rightarrow I_C = 1mA$

$\beta = 200 \Rightarrow I_C = 1,3mA$ $\left(I_C = \frac{9,4}{5 + \frac{5}{200} + \frac{440}{200}} \right)$

POLARIZARE 4R





$E_e = \frac{V_{cc} R_2}{R_1 + R_2} = 5V$

$R_e = R_1 \| R_2 = 2,7K$

P.S.F (point de fonctionnement)

$$\Rightarrow E_c = R_e I_B + U_{BE} + I_E R_4$$

$$E_c = R_e \frac{I_c}{\beta} + U_{BE} + \left(\frac{I_c}{\beta} + I_c\right) R_4$$

$$I_c = \frac{E_c - U_{BE}}{\frac{R_e}{\beta} + \frac{R_4}{\beta} + R_4} = \frac{5 - 0{,}6}{\left(\frac{2{,}2}{100} + \frac{4{,}4}{100} + 4{,}4\right) k} =$$

$$= \frac{4{,}4}{4{,}4} \simeq 1\,mA \qquad (\beta = 100)$$

$$I_c \simeq 1\,mA \qquad (\beta = 200)$$

$$P_S \simeq I_c \cdot U_{CE} \qquad (+ I_B U_{BE})$$

$$U_{CE} = 3V$$

$$I_c = 4mA$$